NOTES

Contents

Rationalizing the denominator

In general, you want to avoid radicals in the denominator. To rationalize the denominator, multiply both the numerator and the denominator by the conjugate of the denominator.

$$
\begin{aligned}
&\frac{1}{1+\sqrt{2}} \\
&= \frac{1}{1+\sqrt{2}} \cdot \frac{1-\sqrt{2}}{1-\sqrt{2}} \\
&= \frac{1-\sqrt{2}}{\left(1+\sqrt{2}\right)\left(1-\sqrt{2}\right)} \\
&= \frac{1-\sqrt{2}}{1^2-\sqrt{2}^2} \\
&= \frac{1-\sqrt{2}}{1-2} \\
&= \frac{1-\sqrt{2}}{-1} \\
&= -1+\sqrt{2}
\end{aligned}
$$

Exercises

1.

$$
\begin{aligned}
&\frac{1+\sqrt{8}}{\sqrt{9}-\sqrt{2}} \\
&= \frac{\left(1+\sqrt{8}\right)\left(\sqrt{9}+\sqrt{2}\right)}{7} \\
&= \frac{\sqrt{9}+3\sqrt{8}+\sqrt{2}+\sqrt{16}}{7} \\
&= \frac{3+6\sqrt{2}+\sqrt{2}+4}{7} \\
&= \frac{7+7\sqrt{2}}{7} \\
&= 1+\sqrt{2}
\end{aligned}
$$

Partial fraction decomposition

We have something of the form:

$$\frac{1}{(x-1)(3x-2)}$$

and we want to decompose it into partial fractions:

$$\frac{1}{(x-1)(3x-2)} = \frac{A}{x-1} + \frac{B}{3x-2}$$

Multiplying both sides by $(x-1)(3x-2)$, we get:

$$1 = A(3x-2) + B(x-1)$$

Setting $x = 1$, we get:

$$1 = A(3(1)-2) + B(1-1) = A$$

Setting $x = \frac{2}{3}$, we get:

$$1 = A\left(3\left(\frac{2}{3}\right) - 2\right) + B\left(\frac{2}{3} - 1\right) = -\frac{B}{3}$$

Solving for B, we get:

$$B = -3$$

So the partial fraction decomposition is:

$$\frac{1}{(x-1)(3x-2)} = \frac{1}{x-1} - \frac{3}{3x-2}$$

Exercises

1. Decompose $\dfrac{44x+48}{(x+6)(x-3)(x+2)}$

$$\frac{44x+48}{(x+6)(x-3)(x+2)} = \frac{A}{x+6} + \frac{B}{x-3} + \frac{C}{x+2}$$
$$44x + 48 = A(x-3)(x+2) + B(x+6)(x+2) + C(x+6)(x-3)$$

Set $x = 3$:

$$\begin{aligned} 44 \cdot 3 + 48 &= B \cdot 9 \cdot 5 \\ \frac{180}{9 \cdot 5} &= B \\ B &= 4 \end{aligned}$$

Set $x = -2$:

$$\begin{aligned} 44 \cdot -2 + 48 &= C \cdot 4 \cdot -5 \\ -40 &= C \cdot -20 \\ C &= 2 \end{aligned}$$

Set $x = -6$:

$$\begin{aligned} 44 \cdot -6 + 48 &= A \cdot -9 \cdot -4 \\ A &= -6 \end{aligned}$$

2. Given that

$$\frac{-2x-9}{x+3}x = \frac{1}{x+3} + \frac{B}{x}$$

find B

$$
\begin{aligned}
\frac{-2x-9}{x+3}x &= \frac{1}{x+3}+\frac{B}{x} \\
(-2x-9) &= x+B(x+3) \\
-3x-9 &= B(x+3) \\
\frac{-3(x+3)}{x+3} &= B \\
B &= -3
\end{aligned}
$$

3. Given that

$$
\begin{aligned}
\frac{-9x+36}{(x-3)(x-6)} &= -\frac{3}{x-3}+\frac{B}{x-6} \\
-9x+36 &= -3(x-6)+B(x-3) \\
-6x+18 &= B(x-3) \\
\frac{-6(x-3)}{x-3} &= B \\
B &= -6
\end{aligned}
$$

Statistics

Variance

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2$$

Covariance

Covariance is the measure of the joint variability of two random variables.

$$\mathrm{Cov}(x,y) = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$$

Graphing

Graphing cubic curves with one distinct real root

If we are given one root of a cubic equation, we can use synthetic division to factor it.

Example

Given that $(x-3)$ is a factor of $f(x) = x^3+x^2-4x-24$, sketch the curve of $y = f(x)$.

First, synthetic division:

	x^3	x^2	x^1	x^0
3	1	1	-4	-24
		3	12	24
	1	4	8	0

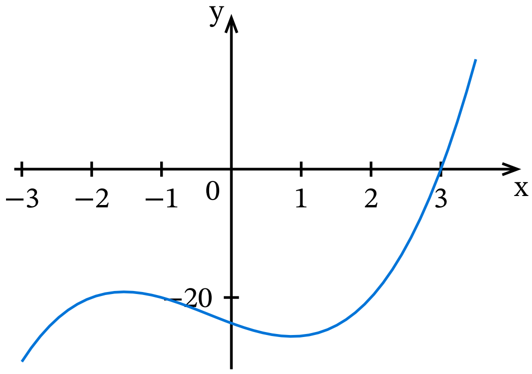
So we have $f(x) = (x-3)\left(x^2+4x+8\right)$

$\therefore x = 3$ is a root

The discriminant of $x^2 + 4x + 8$ is $4^2 - 4 \cdot 1 \cdot 8 = -16$, which is negative, so $x = 3$ is the only real root.

The y intercept is $f(0) = 0^3 + 0^2 - 4 \cdot 0 - 24 = -24$

Now we can draw the curve:



Logarithms

Domain

$(0, \infty)$

Range

$(-\infty, \infty)$

Trigonometry

Unit circle

Reference angle

The *reference angle* of an angle is the smallest angle between the terminal side of the angle and the x-axis. It is always a positive angle less than or equal to 90 degrees.

Calculus

Limits

Limits of logarithmic functions

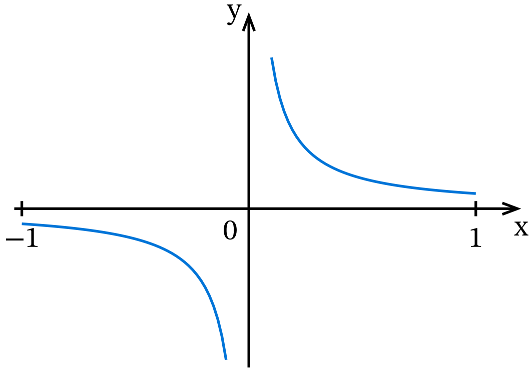
For any real n:

$\lim_{x \to n} \log(x) = \log(x)$

For infinity, remember the domain and apply transformations.

Limits of reciprocal functions

Remember, reciprocal functions have the form $f(x) = \frac{1}{x}$. The limit of a reciprocal function as x approaches infinity is always zero.



In general, for any non-constant polynomial $P(x)$, we have $\lim_{x \to \pm\infty} \frac{1}{P(x)} = 0$

But for $\lim(x \to 0)$, we have to consider the degree of the leading term of $P(x)$.

- Any function of the form $y = \frac{1}{x^n}$ where n is odd has the same shape as $y = \frac{1}{x}$. There is no limit at $x = 0$ because left and right limit are not the same.
- Any function of the form $y = \frac{1}{x^n}$ where n is even has the same shape as $y = \frac{1}{x^2}$. There is a limit, since the function will approach ∞ or $-\infty$.

Differentiation

Constant multiple rule

$$\frac{d}{dx}(kf(x)) = k\frac{d}{dx}(f(x))$$

Sum rule

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

For example for $y = x^2 + x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^2 + x)\\ &= \frac{d}{dx}(x^2) + \frac{d}{dx}(x)\\ &= 2x + 1\end{aligned}$$

The power rule

Given a monomial $f(x) = x^n$, where n is a real number,

$f'(x) = nx^{n-1}$

Integration

The antiderivative

The antiderivative is the opposite of the derivative.

Infinitely many antiderivatives map to the same function.

For example, since the derivative of x^2 is $2x$, we have x^2 as an antiderivative of $2x$. But $x^2 + 2$, $x^2 + 1$, etc. are also antiderivatives of $2x$.

In general, the antiderivatives of $2x$ follow the pattern $x^2 + C$ where C is a constant.

We can also formulate that using the $\int$ symbol for integration:

$$\int 2x dx = x^2 + C$$

Where C is called the **constant of integration**.

These integrals are **indefinite integrals**.

The function being integrated (here $2x$) is called the **integrand**.

The power rule for integration

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Exercises

1. Antiderivative of $\frac{1}{x^4}$?

$$\begin{aligned}\int \frac{1}{x^4} dx &= \int x^{-4} dx \\ &= \frac{x^{-4+1}}{-4+1} + C \\ &= \frac{x^{-3}}{-}3 + C \\ &= -\frac{1}{3}x^3 + C\end{aligned}$$

2. Antiderivative of $x^{\frac{5}{8}}$

$$\begin{aligned}\int x^{\frac{5}{8}} dx &= \frac{x^{\frac{5}{8}+1}}{\frac{5}{8}+1} + C \\ &= \frac{x^{\frac{13}{8}}}{\frac{13}{8}} + C \\ &= \frac{8x^{\frac{13}{8}}}{13} + C\end{aligned}$$

3. Antiderivative of $\sqrt[5]{x^7}$

$$\begin{aligned}\int x^{\frac{7}{5}} dx &= \frac{x^{7/5+1}}{7/5+1} + C \\ &= \frac{x^{12/5}}{12/5} \\ &= \frac{5x^{12/5}}{12}\end{aligned}$$

4. Calculate $\int \left(\frac{z}{5}\, dz\right)$

$$\int\left(\frac{z}{5}dz\right) = \frac{1}{5}\int z\,dz$$
$$= \frac{1}{5}\cdot\left(\frac{z^2}{2}+C\right)$$
$$= \frac{z^2}{10}+C$$

The constant factor rule for indefinite integrals

To take the integral of a power function with a constant factor, like $5x^2$, we can take the constant factor out of the integral:

$$\int 5x^2dx = 5\cdot\int x^2dx$$

This is the **constant factor rule**. In general:

$$\int kf(x)dx = k\int f(x)dx$$

Where k is a **constant**. It does not work if k is a variable.

Exercises

1. Calculate $\int \frac{1}{2y^5}dy$

$$\int \frac{1}{2y^5}dy = \frac{1}{2}\int y^{-5}$$
$$= \frac{1}{2}\cdot\frac{y^{-5+1}}{-5+1}+C$$
$$= \frac{1}{2}\cdot\frac{y^{-4}}{-4}+C$$
$$= \frac{1}{2}\cdot-\frac{1}{4y^4}+C$$
$$= -\frac{1}{8y^4}+C$$

2. Calculate $\int \frac{1}{3x^6}dx$

$$\int \frac{1}{3x^6}dx = \frac{1}{3}\cdot\int x^{-6}$$
$$= \frac{1}{3}\cdot\frac{x^{-5}}{-5}+C$$
$$= \frac{1}{3}\cdot-\frac{1}{5x^5}+C$$
$$= -\frac{1}{15x^5}+C$$

3. Calculate $\int \frac{\pi}{2}dz$

$$
\begin{aligned}
\int \frac{\pi}{2} dz &= \frac{\pi}{2} \int z^0 dz \\
&= \frac{\pi}{2} \cdot z + C \\
&= \frac{\pi z}{2} + C
\end{aligned}
$$